An update to this article is included at the end

Acta Psychologica 260 (2025) 105725

Contents lists available at ScienceDirect

Acta Psychologica

journal homepage: www.elsevier.com/locate/actpsy

Learners' AI dependence and critical thinking: The psychological mechanism of fatigue and the social buffering role of AI literacy

Jinrui Tian [a], Ronghua Zhang [a,*]

[a] *Institute of Developmental and Educational Psychology, School of Marxism, Wuhan University, 430072, Wuhan, China*

ARTICLE INFO

Keywords:
AI dependence
Cognitive fatigue
Information literacy
Critical thinking

ABSTRACT

With the growing integration of artificial intelligence (AI) in education, understanding its cognitive implications has become increasingly important. This study examines how university students' AI dependence influences their critical thinking, exploring cognitive fatigue as a mediating mechanism and information literacy as a moderating factor. Data were collected from 580 Chinese university students, and a moderated mediation model (PROCESS Model 8) was tested. Results indicated that greater AI dependence was associated with lower levels of critical thinking, with cognitive fatigue partially mediating this relationship. Information literacy buffered the negative impact of AI dependence on critical thinking but also amplified cognitive fatigue when AI reliance was high. These findings extend theories of cognitive offloading and automation bias by revealing the dual role of information literacy in AI-supported learning. Practical implications include the need for digital literacy programs and instructional designs that manage cognitive fatigue in AI-integrated education.

1. Introduction

Artificial Intelligence (AI) is transforming various sectors of society, including education, healthcare, and business (Dwivedi et al., 2023; Lo et al., 2024; Lund & Wang, 2023). With the widespread adoption of generative AI technologies—such as ChatGPT—AI has become increasingly embedded in everyday academic practices, enabling students to delegate tasks like summarizing, writing, and even problem-solving (Jagadeesh et al., 2023; King & ChatGPT, 2023).

However, this convenience brings growing concerns about students' cognitive development. Scholars have cautioned that excessive AI use may promote cognitive offloading, reduce critical evaluation, and diminish independent thinking over time (Risko & Gilbert, 2016; Storm et al., 2017). Cognitive offloading refers to the use of external tools to store or process information, thereby conserving internal cognitive resources (Risko & Gilbert, 2016). While such strategies can enhance efficiency, habitual reliance on AI tools may lead to cognitive inertia—a reduced tendency for reflective thinking and analytical reasoning (Mun, 2023).

Emerging empirical studies support these concerns. Students who frequently rely on generative AI tend to demonstrate lower engagement in deep learning and are less likely to apply critical analysis, especially in unfamiliar or ill-structured tasks (Lund & Wang, 2023; Yasmin Khairani Zakaria et al., 2025). A recent meta-analysis revealed that ChatGPT use exerts mixed but generally moderate effects on students' learning performance, perception, and higher-order thinking, highlighting the contextual nature of AI's educational impact (Wang & Fan, 2025). These findings suggest that the influence of AI on learning is not uniformly positive or negative but depends on task type, learner characteristics, and instructional context.

At the same time, concerns are mounting regarding the cognitive costs of AI dependency. Constant cognitive engagement with AI systems can induce cognitive fatigue—a state of mental exhaustion that impairs attention, reasoning, and judgment (Jagadeesh et al., 2023; Zhang et al., 2023). An ongoing randomized controlled experiment further seeks to quantify how generative AI use affects cognitive effort and task performance, providing a rigorous empirical framework for testing cognitive-load-based mechanisms (Chen et al., 2025). Such studies underscore the importance of understanding not only the learning outcomes but also the cognitive processes underlying AI-assisted performance.

Notably, AI may affect learning in both beneficial and detrimental ways depending on how students interact with these systems. Empirical findings indicate that the quality of human–AI interaction and AI output indirectly influences learning outcomes through motivation and self-efficacy, with creativity serving as a moderator (Bai & Wang, 2025). These results imply that high-quality engagement with AI can sustain

* Corresponding author.
E-mail address: ronghuazhang@whu.edu.cn (R. Zhang).

https://doi.org/10.1016/j.actpsy.2025.105725
Received 3 September 2025; Received in revised form 6 October 2025; Accepted 6 October 2025
Available online 11 October 2025
0001-6918/

learning motivation and self-regulation, whereas superficial reliance may foster passivity and cognitive overload.

Beyond cognitive mechanisms, individual differences such as information literacy may further shape how students experience and respond to AI use. Information literacy—the ability to locate, evaluate, and apply information effectively—enables students to assess the credibility of AI outputs, detect biases, and apply critical verification strategies (Liebrenz et al., 2023; Paul et al., 2023). Recent evidence shows that AI literacy moderates the impact of AI use on students' creativity through learning engagement, reinforcing the need to examine literacy-related moderators (Zhou & Peng, 2025). In this sense, information literacy and its AI-specific extensions may serve as cognitive filters that determine whether AI functions as a tool for enhancement or a source of dependency.

These gaps underscore the need for more nuanced investigations into both mediating and moderating mechanisms. Drawing on Social Cognitive Theory (Bandura, 1986) and Cognitive Load Theory (Sweller, 1988), this study examines the impact of AI dependence on students' critical thinking, focusing on the mediating role of cognitive fatigue and the moderating role of information literacy. SCT emphasizes the reciprocal interaction between personal, behavioral, and environmental factors, such as technology use habits and self-regulation (Mun, 2023). CLT explains how overwhelming task demands or information complexity can deplete cognitive resources, thereby impeding higher-order thinking (Sweller, 1988; Zhang et al., 2023).

In this context, the present study addresses two key knowledge gaps: (1) how cognitive fatigue mediates the relationship between AI dependence and critical thinking; and (2) how information literacy moderates these relationships. By testing a moderated mediation model among university students, this research expands current theoretical frameworks and provides empirical insights into how AI affects cognitive functioning in academic settings. Findings may inform the design of AI-supported learning environments that foster rather than hinder students' critical capacities.

1.1. The relationship between AI dependence and critical thinking

The widespread adoption of generative AI tools such as ChatGPT in higher education has sparked increasing interest in their effects on students' cognitive development—particularly critical thinking. While AI offers rapid access to information and support for academic tasks, overdependence on such tools may reduce students' engagement in independent analysis and reflective reasoning (Bahroun et al., 2023; Cezar & Maçada, 2023).

Empirical studies have consistently found that students who heavily rely on AI tools tend to show weaker abilities in evaluating information and solving problems autonomously (Bahrini et al., 2023; Hinss et al., 2022). This pattern of dependence is linked to cognitive inertia, where repeated outsourcing of thinking processes leads to diminished intellectual effort (de la Puente et al., 2024; Lee et al., 2025). Passive AI use—such as copy-pasting content without evaluation—has also been associated with surface-level learning and limited comprehension of complex topics (Dwivedi et al., 2023; Gerlich, 2025).

From a theoretical perspective, cognitive offloading provides a useful explanation for these effects. When students externalize cognitive tasks to AI systems, it reduces internal cognitive demands but may also compromise deeper cognitive processes such as inference, synthesis, and evaluation (Chaudhry et al., 2023; Storm et al., 2017). Over time, habitual offloading may foster "cognitive laziness" and impede the development of higher-order thinking skills. Therefore, we hypothesize that AI dependence is negatively associated with critical thinking (H1), especially when the use of AI is uncritical and habitual.

However, the relationship between AI use and cognition is not uniformly negative. Some studies suggest that AI can enhance critical thinking when used as a scaffold for active learning. For instance, using ChatGPT to support structured debate or to critique its outputs can prompt more analytical reasoning and deeper engagement with content (de la Puente et al., 2024; Supriyanti et al., 2020; Walter, 2024; Yilmaz & Yilmaz, 2023a). These findings imply that the cognitive impact of AI use is moderated by how learners engage with the tool.

Thus, AI is not inherently detrimental to student cognition. When used reflectively and with appropriate regulation, it may serve as a tool for intellectual stimulation. In contrast, excessive or passive AI use may impair the very skills it aims to enhance (Strzelecki, 2023).

Despite growing research on AI's cognitive benefits and drawbacks, there remains a lack of clarity regarding the psychological mechanisms that explain these effects. Specifically, few studies have investigated whether cognitive fatigue mediates the relationship between AI dependence and critical thinking, or whether information literacy moderates this pathway.

To address these questions, the present study employs a moderated mediation model (PROCESS Model 8). Compared to structural equation modeling (SEM), which is ideal for latent variables and large model testing, PROCESS provides a robust and parsimonious tool for examining mediation and moderation with observed variables in a cross-sectional design.

In sum, this study builds on emerging literature to investigate how students' dependence on generative AI affects their critical thinking, with cognitive fatigue as a mediator and information literacy as a moderator. Understanding these mechanisms will help clarify when and how AI use supports or inhibits cognitive engagement in academic settings.

1.2. The mediating role of cognitive fatigue

Cognitive fatigue refers to a state of mental exhaustion characterized by diminished attention, slower cognitive processing, and reduced motivation during sustained mental effort (Hinss et al., 2022). In the context of AI use in education, this form of fatigue may play a critical role in linking excessive AI dependence with impaired critical thinking.

Theoretically, reliance on AI systems can affect cognitive functioning through two opposing yet converging pathways. On one hand, cognitive underload may occur when students delegate substantial portions of thinking to AI tools, leading to insufficient cognitive stimulation. On the other hand, the sheer volume and complexity of AI-generated content may cause information overload, taxing students' mental processing capacity. Both pathways are associated with the onset of cognitive fatigue (Behrens et al., 2023; Lee et al., 2025).

These dual processes are well explained by cognitive offloading theory and information overload theory. According to Risko and Gilbert (2016), individuals tend to shift cognitive effort onto external tools to conserve mental energy—a process that, over time, can reduce deep engagement. Similarly, Storm et al. (2017) showed that habitual external information-seeking leads to greater dependence and weaker internal analytic processing.

Empirical findings also support the link between technological overuse and cognitive fatigue. Hinss et al. (2022) reported that extended human–computer interaction impairs cognitive flexibility and situational awareness. In academic settings, Gerlich (2025) found that students who frequently rely on AI tools tend to process information more superficially, with less sustained analytical effort.

While some neuroscientific studies have linked prolonged cognitive effort with physiological markers of fatigue (e.g., Mao et al., 2022; Wiehler et al., 2022), these findings are preliminary and beyond the scope of this behavioral investigation. Thus, we focus on psychological and behavioral manifestations of fatigue, such as avoidance of cognitively demanding tasks and reliance on heuristics.

For example, Ahmed and Rasul (2023) showed that individuals experiencing digital fatigue were more prone to accept misinformation without critical evaluation. This pattern parallels what may occur in AI-supported learning: students experiencing fatigue may increasingly accept AI outputs without verification, resulting in diminished critical thinking. Such behavior aligns with ego-depletion theory, which posits that cognitive resources are finite and can be depleted through continuous effort, thereby weakening self-regulation (Inzlicht & Schmeichel, 2012).

Taken together, these insights suggest that cognitive fatigue may serve as a key mediator in the relationship between AI dependence and critical thinking. Through a combination of cognitive underload, information overload, and mental resource depletion, frequent reliance on AI may lead to cognitive exhaustion, which in turn impairs students' capacity for reflective and evaluative thinking.

Based on this reasoning, we propose that cognitive fatigue mediates the negative association between AI dependence and critical thinking (H2).

1.3. The moderating role of information literacy ability

Information literacy—the capacity to locate, evaluate, and apply information effectively—may shape how students engage with AI systems and regulate their cognitive effort (Supriyanti et al., 2020; Walter, 2024). Students with stronger literacy skills are more likely to approach AI-generated content with critical scrutiny rather than blind acceptance. This enables them to detect bias, question accuracy, and verify claims, thereby preserving higher-order thinking processes such as analysis and evaluation (Bahroun et al., 2023; Yilmaz & Yilmaz, 2023a).

Research consistently shows that information literacy is positively associated with critical thinking. Instructional programs that improve students' source evaluation and reasoning tend to enhance performance on ill-structured tasks (Supriyanti et al., 2020; Yilmaz & Yilmaz, 2023a). Similarly, students with stronger literacy skills report less reliance on AI tools as substitutes for independent thought, showing lower levels of cognitive inertia even under heavy AI use (Bahrini et al., 2023).

However, information literacy is not without cognitive costs. Highly literate students often engage in vigilant monitoring—checking, cross-referencing, and refining AI outputs—behaviors that can elevate mental workload and lead to subjective fatigue (Tankelevitch et al., 2024). Related work on digital perfectionism indicates that literate users may set excessively high standards for accuracy, which increases verification cycles and contributes to cognitive strain (Etherson et al., 2024). These findings suggest that information literacy may serve as a "double-edged sword": it buffers against cognitive shortcuts but also imposes heavier monitoring demands (Park & Lee, 2021).

Empirical evidence confirms this trade-off. Cezar and Maçada (2023) found that literate professionals are less prone to informational overload, but they also invest more time in detecting inconsistencies. Likewise, Gerlich (2025) and Hinss et al. (2022) report that literate users are more likely to run secondary searches and verify AI-generated content, preserving accuracy but increasing effort. By contrast, students with low literacy skills tend to accept outputs at face value, reducing immediate cognitive load but increasing susceptibility to misinformation (Strzelecki, 2023).

This paradox is mirrored in other domains. During the COVID-19 infodemic, Mao et al. (2022) observed that individuals with higher health literacy were less misinformed but more fatigued—suggesting that critical engagement itself can become mentally taxing. In AI-supported learning, a similar pattern may occur: students with high information literacy may experience less degradation in critical thinking (due to better evaluation skills) but greater cognitive fatigue (due to intensive monitoring effort).

These insights point to a moderating role of information literacy in two distinct pathways. First, it may attenuate the negative association between AI dependence and critical thinking by enabling reflective engagement (H3). Second, it may amplify the positive association between AI dependence and cognitive fatigue due to elevated cognitive monitoring demands (H4).

From an instructional perspective, this dual role implies that cultivating information literacy is necessary but not sufficient. Educators must also teach cognitive management strategies—such as selective verification, batching evaluations, and adopting "good-enough" thresholds—to prevent overextension. By doing so, students can remain active, discerning users of AI without incurring excessive mental strain (Yilmaz & Yilmaz, 2023b).

1.4. Present study

Although the use of generative artificial intelligence (AI) in education has surged, key theoretical and empirical gaps remain regarding its impact on students' higher-order cognition—particularly critical thinking. Most prior research remains conceptual or descriptive, and few empirical studies have systematically examined the psychological processes linking AI dependence to cognitive outcomes (Bahrini et al., 2023; Bahroun et al., 2023). Moreover, findings are mixed: while some report cognitive benefits from strategic AI use (de la Puente et al., 2024), others warn of overuse leading to cognitive inertia and reduced independent reasoning (Dwivedi et al., 2023). These inconsistencies likely stem from methodological limitations, such as reliance on single-wave surveys and the omission of explanatory mechanisms or boundary conditions (Gerlich, 2025; Hinss et al., 2022).

To address these gaps, the present study integrates Social Cognitive Theory (Bandura, 1986) and Cognitive Load Theory (Sweller, 1988) to examine two key processes: cognitive fatigue as a mediating mechanism, and information literacy as a moderating factor. First, we propose that cognitive fatigue transmits the effect of AI overuse on critical thinking. Prior work shows that fatigue impairs deep reasoning and promotes heuristic shortcuts (Ahmed & Rasul, 2023; Wiehler et al., 2022), yet few studies have formally tested this mechanism in educational AI contexts.

Second, we examine whether information literacy moderates both the direct and indirect links. While high literacy can buffer the negative impact of AI on critical thinking by enabling skeptical engagement (Supriyanti et al., 2020; Yilmaz & Yilmaz, 2023a), it may also increase fatigue due to heightened monitoring demands (Tankelevitch et al., 2024). This presents a potential paradox: information literacy may simultaneously protect critical thinking while amplifying cognitive strain.

Accordingly, we propose the following hypotheses:

H1. AI dependence is negatively associated with critical thinking.

H2. Cognitive fatigue mediates the relationship between AI dependence and critical thinking.

H3. Information literacy moderates the direct association between AI dependence and critical thinking, such that the negative effect is weaker when literacy is high.

H4. Information literacy moderates the indirect (fatigue-mediated) pathway from AI dependence to critical thinking, such that the mediating effect is conditional on the level of literacy.

By testing this moderated mediation model (PROCESS Model 8), this study clarifies both the internal mechanism and external boundary conditions under which AI dependence affects student cognition. This work contributes theoretically by bridging SCT and CLT, and practically by informing interventions that balance AI use with literacy and cognitive resilience strategies (Walter, 2024; Yilmaz & Yilmaz, 2023b).

2. Method

2.1. Participants

A total of 580 university students were included in the final analytic sample, drawn from a range of institutions across China, including "985" Project universities, ordinary first-tier universities, and second-tier universities. Participants ranged in age from 18 to 24 years ($M = 19.22$, $SD = 1.96$), with 213 (36.72 %) identifying as male and 367 (63.28 %) as female. In terms of geographical background, 57.22 % were from urban areas and 42.78 % from rural areas.

Data collection took place between March 7 and March 14, 2025, during offline classroom sessions. Participation was anonymous and voluntary, and students were informed that completion would count toward their course attendance credit. This incentive arrangement was approved by the institutional ethics board.

The initial sample consisted of 722 students. First, 22 participants who reported never using AI tools were excluded based on the inclusion criterion that required at least minimal prior AI usage. Then, 120 responses were removed based on a structured data cleaning protocol conducted using SPSS 26.0: (1) Statistical outliers: 43 cases were excluded due to extreme z-scores (±3) on one or more key variables. (2) Uniform response patterns: 51 responses showed low-effort participation, defined as selecting the same Likert option on more than 80 % of items. (3) Missing critical data: 26 cases were excluded due to substantial missing values on primary variables.

Among the final 580 participants, 57.5 % reported using AI tools on a daily basis, while the remaining 42.5 % used them several times a week. In addition, 74 % of respondents indicated that their primary purpose for using AI was to support academic or work-related tasks, reflecting the growing integration of AI into students' learning routines.

Ethical approval for this study was granted by the Institutional Review Board of Wuhan University, School of Humanities and Social Sciences, under the approval number WHU-HSS-IRB2025052. Although course credit was provided as an incentive, students were clearly informed that participation was voluntary and that opting out would not affect their academic standing.

2.2. Measures

Before completing the questionnaires, participants received a brief scenario-based introduction led by the instructor, including real-world examples and functional demonstrations of commonly used AI tools (e.g., ChatGPT, intelligent recommendation systems) in educational and everyday contexts. This introduction aimed to provide participants with a general understanding of AI-related concepts relevant to the study.

After the introduction, paper-based questionnaires were administered on-site during a controlled classroom session. Participants were instructed to complete all items independently based on their personal experiences and perceptions of AI usage. A uniform timeframe was enforced to ensure consistency and data quality across all responses.

2.2.1. AI dependence scale

AI dependence was measured using an adapted version of the Bergen Facebook Addiction Scale (Andreassen et al., 2012), originally developed for social media but later modified by Zhang et al. (2024) for AI-related contexts in academic research. The present study followed Zhang et al.'s adaptation approach by rewording items to reflect AI use (e.g., "I have tried to reduce my use of ChatGPT but have not succeeded"). The scale includes 6 items rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Higher scores indicate greater AI dependence. The scale demonstrated satisfactory internal consistency (Cronbach's α = 0.88).

2.2.2. Cognitive fatigue measurement

Cognitive fatigue was assessed using the Cognitive Fatigue subscale of the Multidimensional Fatigue Inventory-20 (MFI-20) developed by Smets et al. (1995). The subscale contains 4 items (e.g., "I find it difficult to do things that require thinking"), rated on a 5-point Likert scale (1 = not at all true, 5 = completely true). Higher scores indicate greater mental fatigue. This measure has been validated in prior cognitive psychology studies and showed good internal consistency in our sample (Cronbach's α = 0.81).

2.2.3. Information literacy ability scale

Information literacy was measured using the Self-Assessment Tool for Information Technology Application Ability developed by Yan et al. (2018). Although the scale was originally designed to assess information-technology competencies in pre-service teachers, its dimensions—information awareness, retrieval, evaluation, processing, and application—align closely with internationally recognized constructs of information literacy.

The scale includes 15 items (3 per dimension), rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). A sample item is "I can judge the reliability and authority of information." The original study reported high reliability (α = 0.91); in this study, the full scale showed excellent internal consistency (Cronbach's α = 0.96), with subscale αs ranging from 0.83 to 0.90.

2.2.4. Critical thinking scale

Critical thinking was measured using the 17-item Critical Thinking Scale developed by Hou et al. (2022), which is specifically tailored to the Chinese university student population. The scale comprises three dimensions: analytical thinking, open-mindedness, and practical application. Sample items include: "I am clear about my goals before starting a task" and "I like to solve problems step by step." Responses were recorded on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), with higher scores indicating stronger critical thinking ability. In the present study, the internal consistency was high (Cronbach's α = 0.95).

2.3. Analysis method

All statistical analyses were conducted using SPSS 26.0. Descriptive statistics and Pearson correlation analyses were first performed to assess the relationships among the key variables: AI dependence, cognitive fatigue, information literacy, and critical thinking.

To test the hypothesized mediation model (H2), we used PROCESS macro Model 4 (Hayes, 2017), specifying AI dependence as the independent variable (X), cognitive fatigue as the mediator (M), and critical thinking as the outcome variable (Y). To examine the moderated mediation model (H3 and H4), PROCESS Model 8 was employed, with information literacy included as a moderator (W) of both the direct and indirect paths.

All continuous variables were mean-centered to reduce multicollinearity and enhance the interpretability of interaction terms. We set bootstrap resampling to 5000 iterations, and used bias-corrected 95 % confidence intervals to test the significance of indirect and conditional effects. Heteroscedasticity-consistent standard errors (HC3) were applied throughout to ensure robust estimation.

Where significant interaction effects were found, simple slopes analyses were conducted and plotted at low (−1 SD), mean, and high (+1 SD) levels of information literacy to interpret the nature of moderation. Gender and age were included as control variables in all models.

3. Results

3.1. Measurement validity and common method bias

To assess the reliability and construct validity of the measurement instruments, internal consistency coefficients (Cronbach's α), composite reliability (CR), and average variance extracted (AVE) were calculated based on the correlation matrix. The CR values ranged from 0.81 to 0.96, and the AVE values ranged from 0.51 to 0.59, meeting the recommended thresholds of CR > 0.70 and AVE > 0.50 (Fornell & Larcker, 1981), thereby indicating acceptable convergent validity.

Given that the current study employed a regression-based analysis framework (PROCESS macro), confirmatory factor analysis (CFA) was not performed. We acknowledge this as a methodological limitation and recommend future research to apply structural equation modeling for a more rigorous assessment of the measurement model.

To examine potential common method bias, Harman's single-factor test was conducted. An exploratory factor analysis without rotation revealed that the first unrotated factor accounted for 32.89 % of the total variance, which is below the commonly accepted threshold of 40 % (Podsakoff et al., 2003), suggesting that common method bias was not a major concern in this study.

3.2. Descriptive statistics and correlations

Table 1 presents the means, standard deviations, and Pearson zero-order correlations for all study variables ($N = 580$). On average, participants reported moderate levels of AI dependence ($M = 2.86$, $SD = 0.91$) and cognitive fatigue ($M = 3.52$, $SD = 0.68$), relatively high information literacy ($M = 4.90$, $SD = 0.88$), and moderately high critical thinking ($M = 4.94$, $SD = 0.75$).

Consistent with Hypothesis 1, AI dependence was negatively associated with critical thinking ($r = -0.35$, $p < 0.001$) and positively associated with cognitive fatigue ($r = 0.51$, $p < 0.001$). However, its correlation with information literacy was nonsignificant ($r = -0.03$, $p > 0.05$). Cognitive fatigue also showed a significant negative correlation with critical thinking ($r = -0.42$, $p < 0.001$), while its correlation with information literacy was not statistically significant ($r = -0.06$, $p > 0.05$). Notably, information literacy was positively correlated with critical thinking ($r = 0.39$, $p < 0.001$).

These results provide preliminary support for Hypothesis 1 and satisfy the conditions necessary for subsequent mediation (Hypothesis 2) and moderated mediation analyses (Hypotheses 3–4).

3.3. Mediation analysis: the role of cognitive fatigue

Mediation analysis was conducted using Model 4 of the PROCESS macro (see Table 2). Results indicated that AI dependence significantly predicted cognitive fatigue ($\beta = 0.51$, $p < 0.001$), and cognitive fatigue, in turn, negatively predicted critical thinking ($\beta = -0.33$, $p < 0.001$). After including the mediator, the direct effect of AI dependence on critical thinking decreased from $\beta = -0.35$ to $\beta = -0.19$, but remained statistically significant (p < 0.001), suggesting a partial mediation. As shown in Fig. 1, the moderated mediation model is illustrated, while Fig. 2 and Fig. 3 present the simple slope analyses.

The bootstrapped indirect effect was significant (indirect effect = −0.14, 95 % *CI* [−0.22, −0.11]), indicating that cognitive fatigue partially mediates the relationship between AI dependence and critical thinking. These findings support Hypothesis 2.

3.4. Moderated mediation analysis: the role of information literacy

Table 3 summarizes the results of hierarchical regression analyses testing the proposed moderated mediation model. In the first model, AI dependence significantly predicted lower levels of critical thinking ($\beta = -0.34$, $t = -9.58$, $p < 0.001$), while information literacy was positively associated with critical thinking ($\beta = 0.38$, $t = 10.73$, $p < 0.001$). The interaction between AI dependence and information literacy was also significant ($\beta = 0.11$, $t = 3.01$, $p = 0.003$), suggesting that information literacy moderates the effect of AI dependence on critical thinking. As shown in Fig. 2, simple slope analyses revealed that the negative relationship between AI dependence and critical thinking was stronger at

Table 1
Means, standard deviations, and correlation coefficients of variables.

	M	SD	1	2	3	4
1.AI dependence	2.86	0.91	–			
2.Cognitive fatigue	3.52	0.68	0.51***	–		
3.Information literacy skills	4.90	0.88	−0.03	−0.06	–	
4.Critical thinking	4.94	0.75	−0.35***	−0.42***	0.39***	–

*** $p < 0.001$

Table 2
Results of the mediation analysis between AI dependence and critical thinking via cognitive fatigue.

	Cognitive fatigue		Critical thinking		Critical thinking	
	β	t	β	t	β	t
AI dependence	0.51	14.15***	−0.35	−9.03***	−0.19	−4.31
Cognitive fatigue					−0.33	−7.56***
R^2	0.26		0.12		0.20	
F	200.27***		81.49***		73.28***	

*** $p < 0.001$

low levels of information literacy and weaker at high levels, indicating a buffering (protective) effect of information literacy. Fig. 3 demonstrates the moderating effect of information literacy on the relationship between AI dependence and cognitive fatigue. At all levels of AI dependence (low, medium, and high), individuals with lower information literacy report higher levels of cognitive fatigue. However, the slope is steeper for those with weaker information literacy, suggesting that high information literacy mitigates the increase in fatigue associated with rising AI dependence.

The second model examined the predictors of cognitive fatigue. AI dependence was a significant positive predictor ($\beta = 0.69$, $t = 5.64$, $p < 0.001$), and information literacy also showed a small but significant positive association ($\beta = 0.15$, $t = 2.00$, $p = 0.046$). However, the interaction term was significant and negative ($\beta = -0.06$, $t = -2.25$, $p = 0.025$), suggesting that high levels of information literacy attenuate the positive relationship between AI dependence and cognitive fatigue.

In the third model, critical thinking was regressed on all predictors, including cognitive fatigue. As expected, cognitive fatigue significantly predicted lower levels of critical thinking ($\beta = -0.32$, $t = -7.47$, $p < 0.001$). AI dependence remained a negative predictor ($\beta = -0.45$, $t = -3.42$, $p = 0.001$), and the interaction between AI dependence and information literacy remained significant ($\beta = 0.06$, $t = 2.32$, $p = 0.021$). However, the direct effect of information literacy was no longer statistically significant ($\beta = 0.14$, $t = 1.78$, $p = 0.076$) after controlling for cognitive fatigue.

Overall, the final model accounted for 34 % of the variance in critical thinking ($R^2 = 0.34$). These results support Hypotheses 3 and 4: cognitive fatigue mediates the relationship between AI dependence and critical thinking, and information literacy moderates both the direct effect of AI dependence on critical thinking and the indirect effect through cognitive fatigue.

4. Discussion

4.1. The relationship between AI dependence and critical thinking

The present findings demonstrate a robust negative association between AI dependence and critical thinking, consistent with prior work on automation bias, which suggests that over-reliance on intelligent systems can suppress active reasoning and critical engagement (Parasuraman & Riley, 1997; Zhu & Zhang, 2022). In line with cognitive load theory (Sweller, 1988), our results also point to cognitive fatigue as a key mediating mechanism, wherein prolonged AI use appears to deplete mental resources that are essential for complex thinking.

Though speculative, this aligns with physiological research indicating increased frontal-theta activity—a neural marker of cognitive strain—during prolonged interactions with automated systems (Borghini et al., 2014). However, as these physiological correlates were not directly measured in this study, such mechanisms remain tentative and should be explored in future research using behavioral and neurophysiological methods.

Notably, even after accounting for fatigue, the direct negative link between AI dependence and critical thinking remained significant,

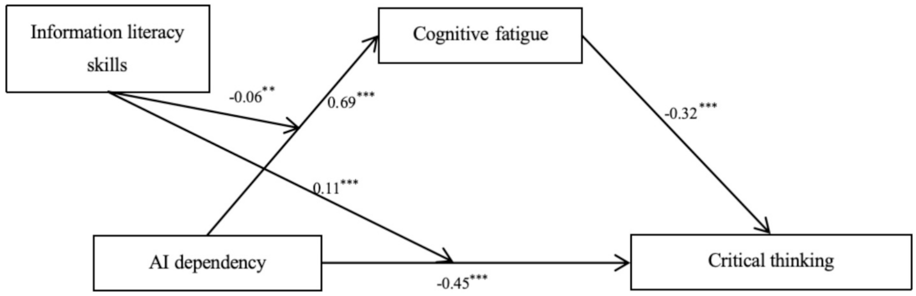

Fig. 1. Moderated mediation model with information literacy moderating the effects of AI dependency on critical thinking via cognitive fatigue.

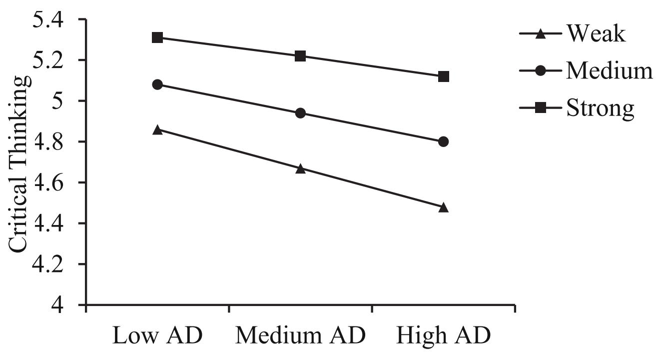

Fig. 2. Simple slopes of AI dependence predicting critical thinking at varying levels of information literacy.

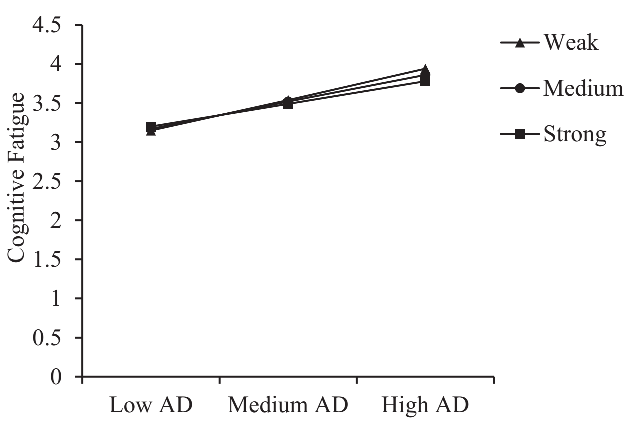

Fig. 3. Simple slopes of AI dependence predicting cognitive fatigue at varying levels of information literacy.

suggesting two partially independent pathways: (1) resource depletion and (2) reduced motivation for analytical thinking. This pattern aligns with theories of cognitive offloading, which posit that the benefits of automation depend on the level of residual monitoring effort that users must still exert (Risko & Gilbert, 2016).

Information literacy emerged as a protective moderator, attenuating the negative impact of AI dependence on critical thinking and weakening the fatigue-inducing effects of AI use. This supports a buffering hypothesis: individuals with higher literacy may engage in more mindful interaction with AI tools, thereby maintaining cognitive engagement and reducing exhaustion. Contrary to earlier interpretation, the data do not suggest that higher literacy exacerbates fatigue; rather, it mitigates it. This clarification aligns with the self-regulation perspective, in which metacognitive awareness can shield users from automation overuse effects (Bawden & Robinson, 2009).

Together, these findings enrich the literature by integrating automation bias, cognitive load, and digital literacy theories into a unified model of how AI usage relates to critical thinking. However, given the cross-sectional design, we caution that these conclusions remain correlational. Longitudinal and experimental studies are needed to clarify directionality and test causality more rigorously.

Table 3
Results of hierarchical regression analyses examining the effects of AI dependence, information literacy, and interaction on critical thinking and cognitive fatigue.

	Critical thinking		Cognitive fatigue		Critical thinking	
	β	t	β	t	β	t
AI dependence	−0.34	−9.58***	0.69	5.64***	−0.45	−3.42***
Information literacy	0.38	10.73***	0.15	2.00*	0.14	1.78
AI dependence × info literacy	0.11	3.01***	−0.06	−2.25**	0.06	2.32*
Cognitive fatigue					−0.32	−7.47***
R^2	0.28		0.27		0.34	
F	74.76***		70.24***		75.37***	

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

4.2. The mediating mechanism of cognitive fatigue between AI dependence and critical thinking

Our findings demonstrate that cognitive fatigue serves as a significant mediator between AI dependence and critical thinking. However, this mediation appears to be partial, suggesting that fatigue is one—rather than the only—mechanism through which AI dependence affects higher-order reasoning.

From a cognitive perspective, fatigue should not be reduced to mere resource depletion; it is multidimensional, dynamically evolving, and tightly connected to failures in metacognitive regulation. The use of large language models often generates lengthy, information-dense outputs, which elevate monitoring, filtering, and verification costs. This aligns with the information overload theory, where increased cognitive burden contributes to reduced vigilance and decision accuracy (Helton & Warm, 2008). Additionally, users who frequently offload search and synthesis to AI may gradually exhibit reduced engagement in discrepancy detection and critical evaluation, consistent with theories of automation bias and "lazy thinking" (Risko & Gilbert, 2016).

While our study did not directly assess physiological states, prior research suggests that fatigue during extended cognitive engagement may be reflected in elevated frontal δ/θ power and decreased prefrontal glucose metabolism—neural markers associated with diminished executive control (Boksem & Tops, 2008). These physiological mechanisms are theoretical inferences, and future studies should incorporate neurocognitive measures to validate their relevance in AI-assisted contexts.

Importantly, our data are consistent with the idea that the relationship between fatigue and critical thinking may be non-linear. That is, when subjective fatigue crosses a certain threshold, individuals are less likely to engage in effortful processing, aligning with the resource depletion model of self-regulation (Inzlicht & Schmeichel, 2012). This could explain why we found partial mediation—below the threshold, cognitive resources may suffice; beyond it, analytical reasoning rapidly deteriorates.

To address this mechanism, several practical strategies may be useful. First, interface design could present AI output in spaced, chunked segments rather than large text blocks, helping reduce instantaneous monitoring load (Warm & Parasuraman, 2009). Second, algorithm design may incorporate explainability cues and uncertainty markers, allowing users to selectively verify only the most ambiguous content (Bawden & Robinson, 2009). Third, pedagogical interventions can train students to recognize signs of fatigue early and switch to low-load tasks as a form of cognitive self-regulation.

Finally, we encourage future longitudinal and experimental studies to examine fatigue thresholds in real time, possibly using diary methods, physiological tracking, or cognitive-behavioral tasks. These could clarify temporal dynamics and provide more robust evidence for the mediating role of fatigue in AI-supported learning environments.

4.3. The moderating effect of information literacy ability on the relationship between AI dependence and critical thinking

Our results show that information literacy moderates both the direct and indirect effects of AI dependence on critical thinking, highlighting its dual role as both a buffer and a risk enhancer.

On the direct path, higher information literacy attenuates the negative relationship between AI dependence and critical thinking ($\beta = 0.11, p = 0.003$). This suggests that literate students are better equipped to evaluate, verify, and synthesize digital content, thereby resisting automation bias and maintaining stronger analytic engagement. This aligns with prior studies emphasizing the protective function of digital and critical literacy (Supriyanti et al., 2020; Yilmaz & Yilmaz, 2023a).

On the indirect path, however, information literacy amplifies the positive relationship between AI dependence and cognitive fatigue ($\beta = -0.06, p = 0.025$). In other words, students with higher literacy levels experience greater fatigue when relying heavily on AI tools. This finding supports the cognitive-monitoring-cost hypothesis (Bawden & Robinson, 2009), which posits that high-verification effort—while improving accuracy—also increases mental load. Although our study did not measure process-level behaviors, prior user-centered research suggests that vigilant AI users often engage in extensive review and revision of machine-generated content, a pattern described as "perfectionist literacy" (Etherson et al., 2024; Tankelevitch et al., 2024). These behaviors, while promoting rigor, can also accelerate cognitive fatigue when monitoring exceeds manageable levels.

Taken together, these results underscore a "double-edged" moderation pattern: information literacy simultaneously shields against passive overreliance on AI and heightens the risk of cognitive overload. To mitigate these trade-offs, literacy instruction should combine evaluative skills with load-management strategies—such as batching verification tasks, using confidence cues to triage content, and adopting "good-enough" standards for low-stakes AI interactions.

For system designers, embedding source transparency, uncertainty markers, and confidence scores can help literate users sustain scrutiny without surpassing their cognitive threshold. These approaches may enable students to maintain critical engagement without incurring unsustainable fatigue.

4.4. Theoretical contributions and practical implications

This study makes two core contributions to both theory and practice. Theoretically, grounded in Cognitive Offloading Theory and Automation Bias Theory and informed by Cognitive Load Theory, it refines and extends existing perspectives on how generative AI shapes human cognition. The findings reveal that reliance on AI does not uniformly reduce cognitive demands; rather, it introduces a paradoxical dynamic in which cognitive fatigue can increase under conditions of high information literacy. This occurs because literate users engage in heightened monitoring, verification, and synthesis to ensure responsible use of AI outputs (Risko & Gilbert, 2016; Tankelevitch et al., 2024).

By identifying a dual-pathway model—one direct (reduced self-initiated processing) and one indirect (fatigue-induced resource depletion)—this study bridges two theoretical traditions: offloading as a load-reduction mechanism and automation bias as a judgmental distortion. Introducing cognitive fatigue as an intervening process provides a unifying account of how external assistance can both conserve and consume cognitive resources, thus refining Cognitive Offloading Theory's assumptions about efficiency gains. Moreover, the discovery of threshold-dependent mediation and double-edged moderation effects establishes critical boundary conditions for AI-mediated cognition: cognitive offloading and fatigue effects are not inherently beneficial or harmful but depend on user traits and cognitive thresholds (Etherson et al., 2024; Wiehler et al., 2022). This contributes to a broader integration between cognitive and socio-cognitive frameworks, illustrating how technological affordances and literacy-based regulation jointly determine higher-order reasoning.

Practically, the findings offer actionable insights for both educators and system designers aiming to foster sustainable critical thinking in AI-rich environments. For educational institutions, information-literacy instruction should go beyond source evaluation and include cognitive load-regulation strategies, such as verifying selectively rather than exhaustively, applying "good-enough" standards for routine tasks, and recognizing early signs of mental fatigue (Bawden & Robinson, 2009). These strategies can help students maintain critical engagement without becoming overwhelmed.

For AI developers and interface designers, systems should be restructured to surface uncertainty flags, provide explainable and layered outputs, and present information in spaced or chunked formats that mitigate instantaneous monitoring load (Warm & Parasuraman, 2009). Such design features can help highly literate users sustain cognitive vigilance while avoiding overload, thereby preserving the benefits of critical scrutiny without incurring excessive mental cost.

Taken together, these theoretical and practical contributions underline the importance of equipping users not only with evaluative literacy but also with fatigue-management capacity. This dual focus is essential for sustaining critical thinking in the age of algorithmic mediation. While our results establish clear statistical associations, future research should employ longitudinal and experimental designs—including behavioral and physiological indicators—to test causal mechanisms and validate the durability of these effects over time.

4.5. Limitations and future research suggestions

Despite its contributions, this study has several limitations that warrant consideration and suggest directions for future research.

First, the sample consisted solely of university students from a single country, which limits generalizability. While this population is relevant to the academic use of generative AI, the underlying cognitive mechanisms identified—such as fatigue-based mediation and literacy-based moderation—may also operate in professional or workplace learning contexts. Future research should therefore test whether these dynamics generalize across cultures, age groups, and occupational domains to enhance external validity.

Second, the research design was cross-sectional, constraining causal inference. Although the hypothesized relationships are theoretically grounded, the temporal order of variables remains untested; longitudinal and experimental studies are thus needed to establish causal directionality and verify temporal stability.

Third, all data were collected via self-report measures. Although validated scales were used, self-reports are vulnerable to social desirability bias and subjective inaccuracies. Future studies should incorporate behavioral indicators (e.g., decision accuracy, reasoning latency) and physiological markers (e.g., EEG, eye-tracking, pupillometry) to triangulate and validate the constructs of cognitive fatigue and critical thinking, thereby reducing common method bias.

Fourth, this study did not conduct confirmatory factor analysis (CFA) or employ structural equation modeling (SEM). While regression-based mediation and moderation (via PROCESS) capture directional associations, they cannot assess measurement validity or structural fit. Applying CFA and SEM in future work would improve construct reliability and theoretical rigor.

Fifth, downstream performance outcomes (e.g., GPA, task accuracy, retention) were not examined. Because cognitive fatigue and offloading may influence both perception and performance, linking these mechanisms to objective academic or workplace outcomes would substantiate their practical significance.

Sixth, the brief scenario-based AI demonstration at the beginning of the study may have unintentionally introduced priming effects that shaped participants' perceptions of AI. Future designs should randomize exposure conditions or statistically control for activation effects.

Seventh, contextual and motivational variables such as academic workload, task complexity, and intrinsic motivation were not measured. These factors may moderate the effects of AI reliance and fatigue across diverse learning environments, and future models should incorporate them.

Eighth, beyond information literacy, individual differences such as digital self-efficacy, metacognitive awareness, and cognitive reflection may further shape users' engagement with AI tools. Including these constructs could yield a more holistic account of individual variability in AI-mediated cognition.

In conclusion, although these limitations constrain certain interpretations, the study advances theoretical understanding of how AI dependence interacts with cognitive fatigue and critical thinking. Future cross-context and cross-population research will be essential to verify the generalizability, causal mechanisms, and long-term implications of these effects.

5. Conclusion

This study explored how AI dependence, cognitive fatigue, and information literacy interact to influence critical thinking among university students. Several key findings emerged.

First, AI dependence was negatively associated with critical thinking, supporting the view that over-reliance on generative AI may dampen self-initiated reasoning processes. Second, cognitive fatigue partially mediated this relationship: individuals who depended more on AI reported greater fatigue, which in turn predicted lower critical-thinking scores. Third, information literacy moderated both paths of the model, but in distinct ways. It attenuated the negative direct effect of AI dependence on critical thinking, suggesting that literate users are better equipped to preserve analytical engagement. However, it amplified the positive association between AI dependence and cognitive fatigue, likely due to the higher cognitive monitoring demands required to scrutinize AI output. This double-edged moderating role of information literacy reveals a nuanced cognitive trade-off: while it enhances protective reasoning, it also raises mental workload.

Compared to prior research, this study refines existing models by introducing a dual-pathway framework that distinguishes between resource-depletion and reasoning-suppression mechanisms. It also extends digital literacy theory by showing that literacy is not universally beneficial; its effects depend on cognitive load and task context. These findings help reconcile inconsistent reports in the literature regarding whether AI use facilitates or hinders deep thinking.

From a practical standpoint, the results suggest that promoting information literacy alone is insufficient. Cognitive load management strategies and fatigue-aware educational design should accompany literacy instruction to ensure sustainable cognitive engagement with AI tools. However, these recommendations are based on theoretical interpretation and should be tested in future empirical work.

CRediT authorship contribution statement

Jinrui Tian: Writing – original draft, Methodology, Conceptualization. **Ronghua Zhang:** Writing – review & editing, Supervision.

Consent for publication

Not applicable.

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the School of Marxism, Wuhan University. Informed consent was obtained from all participants prior to data collection.

Funding

This research received no external funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Not applicable.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

References

Ahmed, S., & Rasul, M. E. (2023). Examining the association between social media fatigue, cognitive ability, narcissism and misinformation sharing: Cross-national evidence from eight countries. *Scientific Reports, 13*, Article 15416. https://doi.org/10.1038/s41598-023-42614-z

Andreassen, C. S., Torsheim, T., Brunborg, G. S., & Pallesen, S. (2012). Development of a Facebook Addiction Scale. *Psychological Reports, 110*(2), 501–517. https://doi.org/10.2466/02.09.18.PR0.110.2.501-517

Bahrini, A., Khamoshifar, M., Abbasimehr, H., Riggs, R. J., Esmaeili, M., Majdabadkohne, R. M., & Pasehvar, M. (2023). ChatGPT: Applications, opportunities, and threats. In *2023 systems and information engineering design symposium (SIEDS)* (pp. 274–279). IEEE. https://doi.org/10.1109/SIEDS55613.2023.10132220.

Bahroun, Z., Anane, C., Ahmed, V., & Zacca, A. (2023). Transforming education: A comprehensive review of generative artificial intelligence in educational settings through bibliometric and content analysis. *Sustainability, 15*(17), Article 12983. https://doi.org/10.3390/su151712983

Bai, Y., & Wang, S. (2025). Impact of generative AI interaction and output quality on university students' learning outcomes: A technology-mediated and motivation-driven approach. *Scientific Reports, 15*, Article 24054. https://doi.org/10.1038/s41598-025-08697-6

Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice-Hall.

Bawden, D., & Robinson, L. (2009). The dark side of information: Overload, anxiety and other paradoxes and pathologies. *Journal of Information Science, 35*(2), 180–191. https://doi.org/10.1177/0165551508095781

Behrens, M., Gube, M., Chaabene, H., et al. (2023). Fatigue and human performance: An updated framework. *Sports Medicine, 53*(1), 7–31.
Boksem, M. A. S., & Tops, M. (2008). Mental fatigue: Costs and benefits. *Brain Research Reviews, 59*(1), 125–139. https://doi.org/10.1016/j.brainresrev.2008.07.001
Borghini, G., Astolfi, L., Vecchiato, G., Mattia, D., & Babiloni, F. (2014). Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness. *Neuroscience & Biobehavioral Reviews, 44*, 58–75. https://doi.org/10.1016/j.neubiorev.2012.05.003
Cezar, B. G. S., & Maçada, A. C. G. (2023). Cognitive overload, anxiety, cognitive fatigue, avoidance behavior and data literacy in Big Data environments. *Information Processing & Management, 60*(2), Article 103482. https://doi.org/10.1016/j.ipm.2023.103482
Chaudhry, I. S., Sarwary, S. A. M., El Refae, G. A., & Chabchoub, H. (2023). Time to revisit existing student performance evaluation approaches in higher education in a new era of ChatGPT—A case study. *Cogent Education, 10*(1), Article 2210461. https://doi.org/10.1080/2331186X.2023.2210461
Chen, Y., Wang, Y., Wüstenberg, T., Kizilcec, R. F., Fan, Y., Li, Y., … Bärnighausen, T. (2025). Effects of generative artificial intelligence on cognitive effort and task performance: Study protocol for a randomized controlled experiment among college students. *Trials, 26*(1), 244. https://doi.org/10.1186/s13063-025-08950-3
de la Puente, M., Torres, J., Blanco Troncoso, A. L., Hernández Meza, Y. Y., & Márquez Carrascal, J. X. (2024). Investigating the use of ChatGPT as a tool for enhancing critical thinking and argumentation skills in international relations debates among undergraduate students. *Smart Learning Environments, 11*, 55. https://doi.org/10.1186/s40561-024-00347-0
Dwivedi, Y. K., Kshetri, N., Hughes, D. L., Coombs, C., Edwards, J. S., Shareef, M. A., … Kar, A. K. (2023). "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, Article 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642
Etherson, M. E., Hill, A. P., Grugan, M., Madigan, D. J., & Smith, M. M. (2024). Development and validation of the perfectionism literacy questionnaire. *Journal of Psychoeducational Assessment*. https://doi.org/10.1177/07342829241313089
Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50. https://doi.org/10.2307/3151312
Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. *Societies, 15*(1), Article 6. https://doi.org/10.3390/soc15010006
Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (2nd ed.). New York: The Guilford Press.
Helton, W. S., & Warm, J. S. (2008). Signal salience and the mindlessness theory of vigilance. *Acta Psychologica, 129*(1), 18–25. https://doi.org/10.1016/j.actpsy.2008.04.002
Hinss, M. F., Brock, A. M., & Roy, R. N. (2022). Cognitive effects of prolonged continuous human–machine interaction: The case for mental state-based adaptive interfaces. *Frontiers in Neuroergonomics, 3*, Article 898745. https://doi.org/10.3389/fnrgo.2022.898745
Hou, Y., Li, Q., & Li, H. (2022). The structure of Chinese critical thinking and the development of its measurement scale. *Journal of Peking University (Natural Science), 58*(2), 383–390. https://doi.org/10.13209/j.0479-8023.2022.001
Inzlicht, M., & Schmeichel, B. J. (2012). What is ego depletion? Toward a mechanistic revision of the resource model of self-control. *Perspectives on Psychological Science, 7*(5), 450–463. https://doi.org/10.1177/1745691612454134
Jagadeesh, M., Pa, A. S., Ali, A. S., & B S, A. (2023). ChatGPT and its double-edged impact on higher education and research - A detailed survey. In *2023 6th international conference on recent trends in advance computing (ICRTAC), Chennai, India* (pp. 10–17). https://doi.org/10.1109/ICRTAC59277.2023.10480767
King, M. R., & ChatGPT. (2023). A conversation on artificial intelligence, chatbots, and plagiarism in higher education. *Cell Reports Physical Science, 4*(3), Article 101135. https://doi.org/10.1016/j.xcrp.2023.101135
Lee, H.-P., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI conference on human factors in computing systems (paper no. 3713778)* (pp. 1–23). ACM. https://doi.org/10.1145/3706598.3713778.
Liebrenz, M., Schleifer, R., Buadze, A., Bhugra, D., & Smith, A. (2023). Generating scholarly content with ChatGPT: Ethical challenges for medical publishing. *The Lancet Digital Health, 5*(3), e105–e106. https://doi.org/10.1016/S2589-7500(23)00019-5
Lo, C. K., Yu, P. L. H., Xu, S., et al. (2024). Exploring the application of ChatGPT in ESL/EFL education and related research issues: A systematic review of empirical studies. *Smart Learning Environments, 11*, Article 50. https://doi.org/10.1186/s40561-024-00342-5
Lund, B. D., & Wang, T. (2023). Chatting about ChatGPT: How may AI and GPT impact academia and libraries?. In *Library Hi Tech News*. https://doi.org/10.2139/ssrn.4333415. Available at SSRN: https://ssrn.com/abstract=4333415.
Mao, B., Jia, X., & Huang, Q. (2022). How do information overload and message fatigue reduce information processing in the era of COVID-19? An ability–motivation approach. *Journal of Information Science*. https://doi.org/10.1177/01655515221114322 (Advance online publication).
Mun, H. (2023). The impact of digital tool dependency on students' problem-solving abilities: A cognitive load perspective. *Journal of Educational Technology Development and Exchange, 16*(1), Article 3. https://doi.org/10.18785/jetde.1601.03
Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors, 39*(2), 230–253. https://doi.org/10.1518/001872097778543886
Park, Y., & Lee, H. (2021). The double-edged sword of digital literacy: How it helps and hinders student learning in AI environments. *British Journal of Educational Technology, 52*(5), 1457–1474. https://doi.org/10.1111/bjet.13139
Paul, J., Ahmad, N., Wei, S., & Geng, J. (2023). Generative AI, ChatGPT, and marketing: A research agenda. *Psychology & Marketing, 40*(11), 2239–2256. https://doi.org/10.1002/mar.21954
Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879
Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences, 20*(9), 676–688. https://doi.org/10.1016/j.tics.2016.07.002
Smets, E. M. A., Garssen, B., Bonke, B., & De Haes, J. C. J. M. (1995). The multidimensional fatigue inventory (MFI) psychometric qualities of an instrument to assess fatigue. *Journal of Psychosomatic Research, 39*(5), 315–325.
Storm, B. C., Stone, S. M., & Benjamin, A. S. (2017). Using the Internet to access information inflates future use of the Internet to access other information. *Memory, 25*(6), 717–723. https://doi.org/10.1080/09658211.2016.1210171
Strzelecki, A. (2023). To use or not to use ChatGPT in higher education? A study of students' acceptance and use of technology. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2023.2209881 (Advance online publication).
Supriyanti, S., Permanasari, A., & Khoerunnisa, F. (2020). Correlation between information literacy and critical thinking enhancement through PjBL-information literacy learning model. *Journal of Educational Sciences, 4*(4), 774–784. https://doi.org/10.31258/jes.4.4.p.774-784
Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science, 12*(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4
Tankelevitch, L., Kewenig, V., Simkute, A., Scott, A., Sarkar, A., Sellen, A., & Rintel, S. (2024). The metacognitive demands and opportunities of generative AI. In *Proceedings of CHI '24 (article 680)*. https://doi.org/10.1145/3613904.3642902
Walter, Y. (2024). Embracing the future of artificial intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education, 21*(1), 15. https://doi.org/10.1186/s41239-024-00448-3
Wang, J., & Fan, W. (2025). The effect of ChatGPT on students' learning performance, learning perception, and higher-order thinking: Insights from a meta-analysis. *Humanities and Social Sciences Communications, 12*, 621. https://doi.org/10.1057/s41599-025-04787-y
Warm, J. S., & Parasuraman, R. (2009). Vigilance and target detection. In M. D. Gendalman, & J. S. Warm (Eds.), *Cognitive vigilance* (pp. 115–141). Erlbaum.
Wiehler, A., Branzoli, F., Adanyeguh, I., Mochel, F., & Pessiglione, M. (2022). A neuro-metabolic account of why daylong cognitive work alters the control of economic decisions. *Current Biology, 32*(16), 3564–3575.e5. https://doi.org/10.1016/j.cub.2022.07.010
Yan, H.-B., Li, X.-Y., & Ren, Y.-Q. (2018). Development and validation of the self-assessment tool for pre-service teachers' information technology application ability. *e-Education Research*, (1), 98–106. In Chinese: 闫寒冰, 李笑樱, 任友群. (2018). 师范生信息技术应用能力自评工具的开发与验证. 《电化教育研究》.
Yasmin Khairani Zakaria, N., Hashim, H., & Azhar Jamaludin, K. (2025). Exploring the impact of AI on critical thinking development in ESL: A systematic literature review. *Arab World English Journal.*
Yilmaz, R., & Yilmaz, F. G. K. (2023a). Augmented intelligence in programming learning: Examining student views on the use of ChatGPT for programming learning. *Computers in Human Behavior: Artificial Humans, 1*(2), Article 100005. https://doi.org/10.1016/j.chbah.2023.100005
Yilmaz, R., & Yilmaz, F. G. K. (2023b). The effect of generative artificial intelligence (AI)-based tool use on students' computational thinking skills, programming self-efficacy and motivation. *Computers and Education: Artificial Intelligence, 4*, Article 100147. https://doi.org/10.1016/j.caeai.2023.100147
Zhang, S., Zhao, X., Zhou, T., & Kim, J. H. (2024). Do you have AI dependency? The roles of academic self-efficacy, academic stress, and performance expectations on problematic AI usage behavior. *International Journal of Educational Technology in Higher Education, 21*(1), 34.
Zhang, Y., Sun, H., & Chen, L. (2023). Rethinking AI-enhanced education: Potential and pitfalls. *Computers and Education: Artificial Intelligence, 4*, Article 100126. https://doi.org/10.1016/j.caeai.2023.100126
Zhou, M., & Peng, S. (2025). The usage of AI in teaching and students' creativity: The mediating role of learning engagement and the moderating role of AI literacy. *Behavioral Sciences (Basel, Switzerland), 15*(5), 587. https://doi.org/10.3390/bs15050587
Zhu, Y., & Zhang, X. (2022). Relying on AI: How artificial intelligence affects human judgment in collaborative decision-making. *Computers in Human Behavior, 134*, Article 107320. https://doi.org/10.1016/j.chb.2022.107320

Update

Acta Psychologica
Volume 267, Issue , July 2026, Page

DOI: https://doi.org/10.1016/j.actpsy.2026.107179

Acta Psychologica 267 (2026) 107179

Contents lists available at ScienceDirect

Acta Psychologica

journal homepage: www.elsevier.com/locate/actpsy

Corrigendum

Corrigendum to "Learners' AI dependence and critical thinking: The psychological mechanism of fatigue and the social buffering role of AI literacy" [Acta Psychologica 260 (2025) 105725]

Jinrui Tian, Ronghua Zhang *

Institute of Developmental and Educational Psychology, School of Marxism, Wuhan University, 430072, Wuhan, China

The authors regret that two reference entries and three related background citation locations were incorrectly presented due to reference-management and DOI-matching errors during the preparation and final formatting of the reference list. In the affected reference entries, valid DOI records were incorrectly matched with wrong bibliographic information.

First, the citation "Mun, 2023" appeared in two places in the article.

In the first instance, the following sentence was published:

"While such strategies can enhance efficiency, habitual reliance on AI tools may lead to cognitive inertia—a reduced tendency for reflective thinking and analytical reasoning (Mun, 2023)."

This sentence should be corrected as follows:

"While such strategies can enhance efficiency, habitual reliance on AI tools may reduce learners' engagement in reflective thinking and analytical reasoning by encouraging cognitive offloading rather than active problem solving (Gerlich, 2025)."

Accordingly, the reference entry for Mun (2023) should be removed and replaced with:

Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. Societies, 15(1), Article 6. https://doi.org/10.3390/soc15010006

In the second instance, the following sentence was published:

"SCT emphasizes the reciprocal interaction between personal, behavioral, and environmental factors, such as technology use habits and self-regulation (Mun, 2023)."

This sentence should be corrected as follows:

"SCT emphasizes the reciprocal interaction between personal, behavioral, and environmental factors, such as technology use habits and self-regulation (Bandura, 1986)."

Second, the citation "Paul et al., 2023" appeared in the following sentence:

"Information literacy—the ability to locate, evaluate, and apply information effectively—enables students to assess the credibility of AI outputs, detect biases, and apply critical verification strategies (Liebrenz et al., 2023; Paul et al., 2023)."

This sentence should be corrected as follows:

"Information literacy—the ability to locate, evaluate, and apply information effectively—enables students to assess the credibility of AI outputs, identify potential biases, and apply critical verification strategies when using generative AI systems (Liebrenz et al., 2023; Park, 2025)."

Accordingly, the reference entry for Paul et al. (2023) should be removed and replaced with:

Park, J. (2025). A systematic literature review of generative artificial intelligence (GenAI) literacy in schools. Computers and Education: Artificial Intelligence, 8, Article 100,487.

https://doi.org/10.1016/j.caeai.2025.100487

These corrections concern only background citations and reference entries. They do not affect the study design, data, analyses, results, interpretation, or conclusions of the article.

The authors apologize for these errors and for any inconvenience caused.

DOI of original article: https://doi.org/10.1016/j.actpsy.2025.105725.

* Corresponding author.

E-mail address: ronghuazhang@whu.edu.cn (R. Zhang).

https://doi.org/10.1016/j.actpsy.2026.107179

Available online 1 June 2026